European Commission

PRODUCT USER MANUAL

For Arctic Ocean Reprocessed Sea Ice Thickness Products

SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013

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Copernicus Marine Service

Copernicus Europe's eyes on Earth

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2019/04/12	All	Creation of the document	C. Wettre	C. Derval
1.1	2023/09/29	All	Update to data record version 3.0	S. Hendricks	
2.0	2024/05/28	All	Update to data record version 4.0	S. Hendricks	C. Wettre

Table of Contents

RECORD TABLE

GLOSSARY AND ABBREVIATIONS *4*

DATA ACCESS **6**

I INTRODUCTION *7*

I.1 Summary **7**

I.2 History of changes **8**

II DESCRIPTION OF THE PRODUCT SPECIFICATION *9*

II.1 General Information **9**

II.2 Details of datasets **10**

II.3 Additional Information on parameters **10**

II.4 Production System Description **11**

II.4.1 SIT CDR Production System at AWI 12

II.4.2 Observational Data 14

II.5 Grid **15**

II.6 Vertical Levels **16**

II.7 Processing Information **17**

II.7.1 Update Time 17

II.7.2 Temporal extent of analysis and forecast stored on delivery mechanism 17

II.7.3 Time averaging 17

III FILE FORMAT *18*

IV FILES NOMENCLATURE *19*

IV.1 Nomenclature of files when downloaded through the CMEMS Web Portal Subset Service ..... Erreur ! Signet non défini.

IV.2 Nomenclature of original files ..... Erreur ! Signet non défini.

IV.3 Other information: mean centre of Products, land mask value, missing value **20**

IV.4 File size **20**

IV.5 Structure of files **20**

V Annex *21*

GLOSSARY AND ABBREVIATIONS

TAC	Thematic Assembly Centre
SI	Sea Ice
NetCDF	Network Common Data Form
C3S	Copernicus Climate Change Service
CF	Climate Forecast (convention for NetCDF)
CDR	Climate Data Record
FRB	Freeboard
CIS	Central Information System
DGF	Direct Get File. A CMEMS download method. "Direct download" directly transfers the file as stored on the server. It is able to deliver compressed data and should be used if you wish to download larger datasets.
DU	Dissemination Unit. A Dissemination Unit is a kind of Production Unit, with the ability to disseminate its own products or other Production Unit's products to the rest of CMEMS Production Centres and to external Users.
DT	Delayed Time
EC	European Commission
ECMWF	European Centre for Medium Range Weather Forecast
SAR	Synthetic Aperture Radar
SARin	Synthetic Aperture Radar Interferometric Mode
SIC	SIC Sea Ice Concentration
SIT	Sea Ice Thickness
SIType	Sea Ice Type (first-year / multi-year sea ice)

L1B	Level 1B data: Calibrated sensors data from instruments
L1P	Preprocessed Level-1B data
L2	Level 2 data: ground segment geophysical measures, interpreted, calibrated
L2I	Level-2 intermediate: Level-2 data including part of the Level-1 data for the same coverage of the source Level-1 data file.
L3	Level 3 data: Temporally aggregated data on grid.
PC	Production Centre
PU	Production Unit
Directgetfile	CMEMS service tool (FTP like) to download a NetCDF file

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

This guide describes the Arctic reprocessed Sea Ice Thickness data product files from the Sea Ice Thematic Assembly Centre (SI TAC), what data services are available to access them, and how to use the files and services.

The product (SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013) is produced on behalf of the Copernicus Climate Change Services (C3S) and redistributed by CMEMS. It contains monthly Level 3 files in the Northern Hemisphere from October to April with a Climate Data Record (CDR) of reprocessed sea ice thickness version 4 based on radar altimeter data, which consists of data from ERS-2 (Oct. 1995 – Apr 2003), Envisat time series (Oct 2002 - Oct 2010), the CryoSat-2 (Nov 2010 - Apr 2024), Sentinel-3A (Oct 2016 – Apr 2024) and Sentinel-3B (Oct 2018 – April 2024).

A key difference between ERS-2, Envisat and the Sentinel-3 constellation and CryoSat-2 is the orbit coverage. Only CryoSat-2 has a data coverage until 88N, all other altimeters data coverage is limited to latitude lower then 81.5N. Another separation is the type of radar altimeter sensor. The legacy missions ERS-2 and Envisat where equipped with a pulse-limited radar altimeter, while CryoSat-2 and Sentinel-3 platforms feature a synthetic aperture radar altimeter, with superior spatial resolution and sea ice thickness accuracy.

Sea ice thickness retrieval with radar altimeter requires a monthly period to aggregate data for a grid with 25 km resolution and is only feasible in the winter period from October – April.

This data product is redistributed, therefore the full technical documentation including the Algorithm Theoretical Basis Document [RD.1], Product User Guide & Specifications [RD.2], System Quality Assurance Document [RD.3] and Product Quality Assessment Document [RD.4] are available at the C3S Climate Data Store (CDS)[1]. This document provides a summary for users of the Copernicus Marine Service.

The data are freely available under the CMEMS data license. Detailed information on the systems and products are on the CMEMS web site: marine.copernicus.eu.

[1] https://cds.climate.copernicus.eu/cdsapp#!/dataset/satellite-sea-ice-thickness (last accessed: September 2023)

I.2 History of changes

Date	Description of changes and impacted product
2023/09/29	**UPDATE TO NEW VERSION 3.0 OF THE REPROCESSED SEA ICE THICKNESS PRODUCT. VERSION 3.0 IMPROVES SPATIAL COVERAGE DUE TO IMPROVED HANDLING OF RADAR ALTIMETER DATA OF THIN SEA ICE AND IN THE MARGINAL ICE ZONE.**
2024/05/28	**UPDATE TO VERSION 3.0 OF THE REPROCESSED SEA ICE THICKNESS PRODUCT. VERSION 3.0 ADD RADAR ALTIMETER DATA FROM ERS-2, SENTINEL-3A AND SENTINEL-3B AND EXTENDS THE TEMPORAL DATA COVERAGE.**

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Lines	SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013
Geographical coverage	Northern hemisphere
Variables	Sea Ice Thickness
Product Type	Multi Year
Available time series	1995/10/01 – 2024/04/30 (only between October and April)
Temporal resolution	Monthly
Target delivery time	N/A
Delivery mechanism	Marine Data Store
Horizontal resolution	25.0 km grid spacing
Number of vertical levels	1
Format	NetCDF-4, Climate & Forecast (CF) v1.7

Table 1: Details of reprocessed Arctic Sea Ice Thickness (SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013)

II.2 Details of datasets

SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013
Dataset: **c3s_obs-si_glo_phy_my_nh-l3_P1M** **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
sea_ice_thickness [m] Sea Ice Thickness sea_ice_thickness
uncertainty [m] Sea Ice Thickness Uncertainty sea_ice_thickness standard_error
status_flag [1] Sea Ice Thickness Status Flag status_flag
quality_flag [1] Sea Ice Thickness Quality Flag quality_flag

Table 2: list of the datasets and variable names and unit for the SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013 product

II.3 Additional Information on parameters

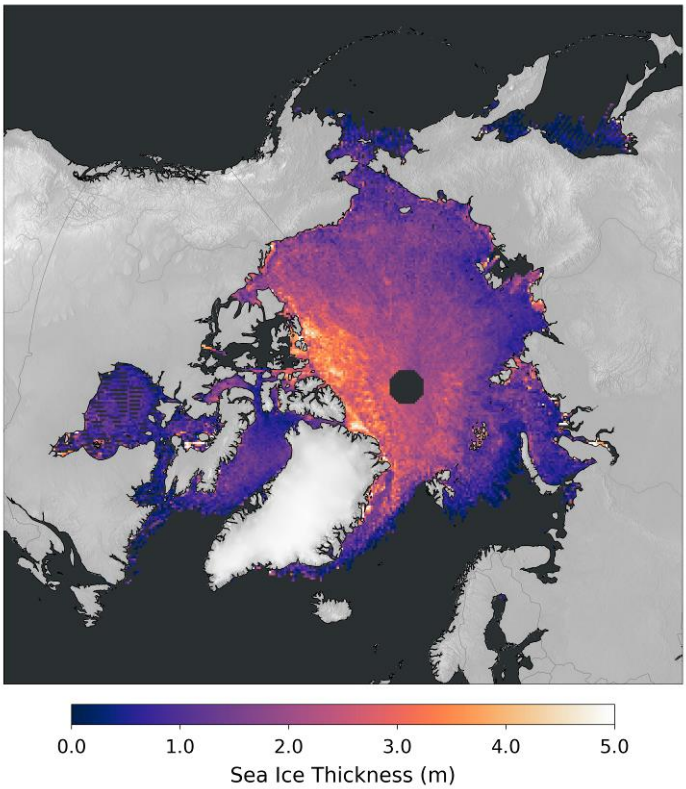
sea_ice_thickness [m]	Defined as mean sea ice thickness of ice-covered area fraction of the grid cell.
uncertainty [m]	Algorithm uncertainty of gridded data.
status_flag [1]	A flag indicating the status of the retrieval: 0: Nominal retrieval 1: No data 2: Open ocean 3: Satellite pole hole 4: Land, lake, or land ice 5: Retrieval failed
quality_flag [1]	A flag indictating an expert estimation of the retrieval : 0: Nominal quality 1: Intermediate quality 2: Low quality 3: No data

II.4 Production System Description

Summary of the operational model setups: Domain

Resolution and grid: 25 km Equal-Area Scalable Earth Grid in version 2

Geographic coverage: Northern hemisphere: North of 55°N



Example of reprocessed sea ice thickness from April 2020.

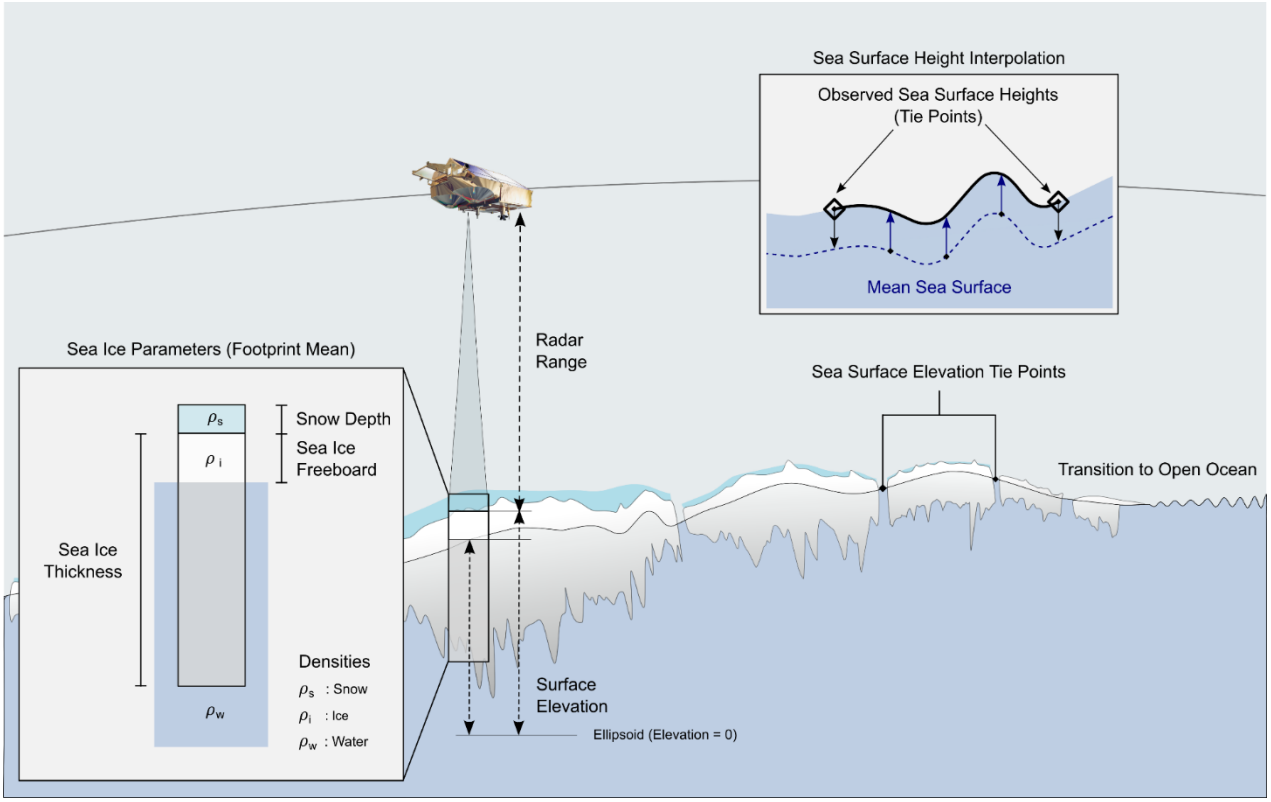
Bathymetry	Surface only
Vertical grid	Surface only
Model Version	V4.0
Production Unit	Alfred Wegener Institute, Germany
Period of validation	1995/10 – 2020/04 (October through April only)
Assimilation scheme	N/A
Assimilated observations	ERS RA, Envisat RA-2, CryoSat-2 SIRAL, Sentinel-3A SRAL, Sentinel-3B SRAL

II.4.1 SIT CDR Production System at AWI

The sea ice thickness production system is hosted at the Alfred Wegener Institute and uses Level-1B orbit data from satellite radar altimeters as well as auxiliary data files to create monthly gridded sea ice thickness products.

The principle of sea ice thickness retrieval using satellite radar altimetry is the estimation of freeboard along the ground track of the satellite (Figure 1). Freeboard is derived as the difference between the surface elevations of sea ice and the sea surface, which are based on radar range measurements over both surface types. This step involves the interpolation of discrete sea surface elevation observations (tie points) in leads between ice floes. The interpolation is aided by a mean sea surface. Sea ice freeboard is then converted into sea ice thickness based on Archimedes Principle, considering snow depth and the densities of snow, sea ice and sea water. Sea ice thickness from all orbits within a month is then aggregated for the SEAICE_GLO_PHY_CLIMATE_L3_MY_011_013 product.

Figure 1: Principle of sea ice freeboard and thickness retrieval using satellite radar altimetry.



Several processing steps with defined input and output processing level are required, which are summarized in Table 1 and the respective algorithm elements are illustrated in Figure 2.

Table 1: Processing steps of reprocessed sea ice thickness production system.

Processor	Content	Input	Output
Level-1 Pre-Processor	• Subsetting and orbit segment merging • Harmonization of different primary data sets	Primary (RA L1b)	L1P
Level-2 Processor	• geophysical variables at full sensor resolution • uncertainties • filtering	L1P	L2i
Level-3 Processor	• Averaging on output grid and grid cell statistics • Application of masks • Status flag	L2i	L3S

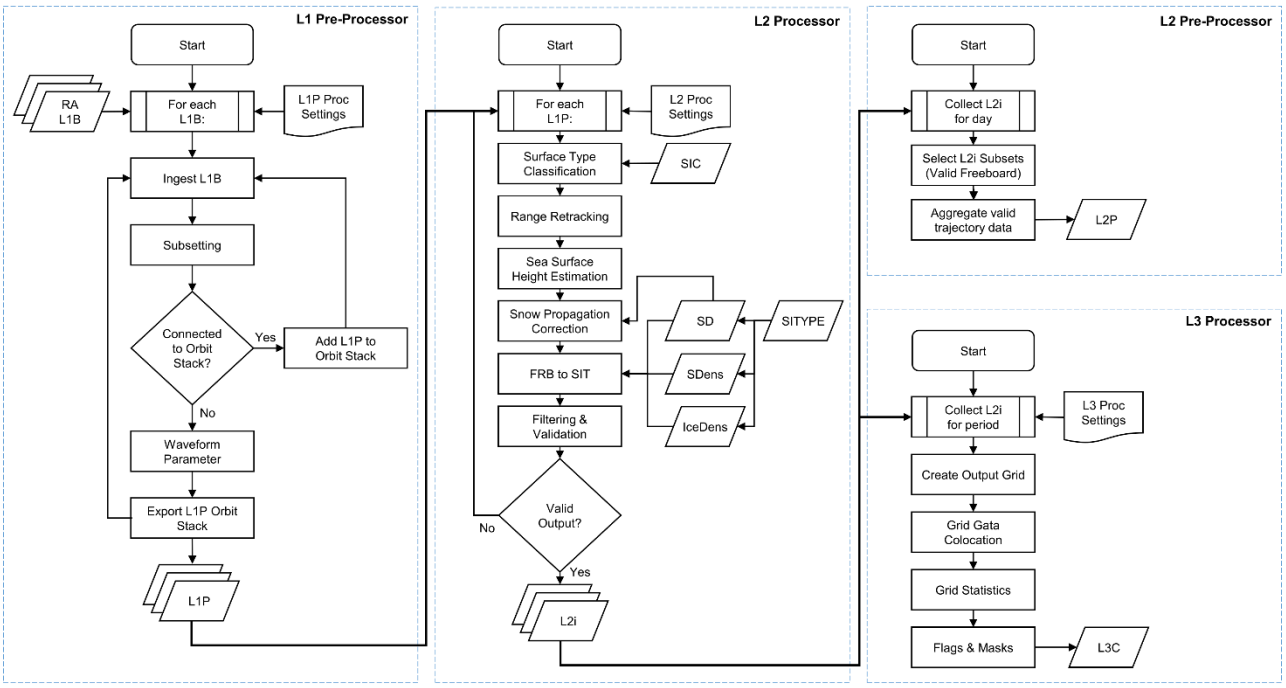


Figure 2: Algorithm workflow of the reprocessed sea ice thickness production system.

The production system is based on the open source python sea ice radar altimetry toolkit (pysiral)[2].

A full description of the algorithm is available in the Algorithm Theoretical Basis Document [RD.1] and a full description of the production system in the System Quality Assurance Document [RD.3].

[2] https://github.com/pysiral/pysiral (last access September 2023)

II.4.2 Observational Data

Input data for the sea ice thickness production system falls into two categories :

1. Primary observations : calibrated radar altimeter sensor data (Table 2)
2. Auxiliary data : geophysical variables for the freeboard retrieval and freeboard to thickness conversion (Table 3).

An detailed description of the data sets is given in the Algorithm Theoretical Basis Document [RD.1].

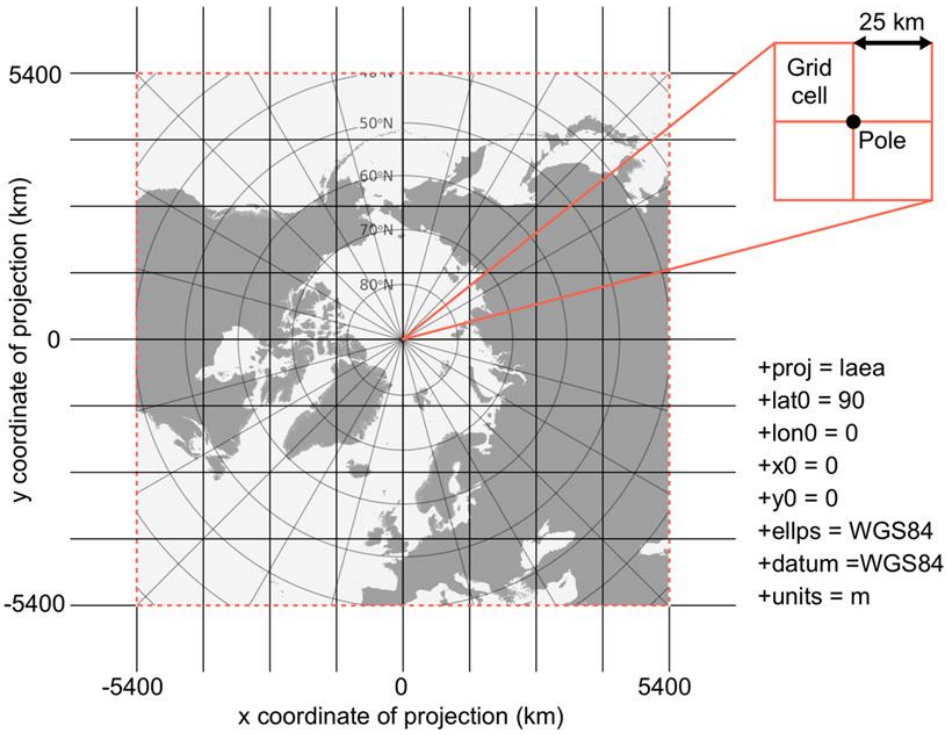
Table 2: Primary observational data products for the sea ice thickness CDR

Platform	Sensor	Source	Processing Level	Coverage
ERS-2	RA	ESA	Sensor Geophysical Data Record (SGDR) (v1.08)	Northern hemisphere (50N < lat < 81.45N) 1995/10 – 2003/04
Envisat	RA-2	ESA	Sensor Geophysical Data Record (SGDR) (v3.0)	Northern hemisphere (50N < lat < 81.45N) 2002/10 – 2012/03
CryoSat-2	SIRAL	ESA	Level-1B (SAR/SARin) Baseline-E	Northern hemisphere (50N < lat < 88N) 2010/11 – 2020/04
Sentinel-3A	SRAL	Copernicus	Level-2 Thematic Sea Ice Product (L2_LAN_SI) Baseline: 005	Global (45N < lat < 81.5N) 2016/04 – ongoing
Sentinel-3B	SRAL	Copernicus	Level-2 Thematic Sea Ice Product (L2_LAN_SI) Baseline: 005	Global (45N < lat < 81.5N) 2018/06 – ongoing

Table 3: Summary of auxiliary data sources, and key characteristics, for the sea ice thickness production system

Auxiliary Type	Description	Temporal Resolution	Coverage	Used for
Mean Sea Surface Height (MSS)	DTU21: Gridded mean sea surface height from multi-sensor altimetry data	Static	Global Oceans	Sea surface height interpolation
Sea Ice Concentration (SIC)	EUMETSAT OSI-SAF SIC CDR v3.0 (OSI-450)	daily	Northern hemisphere	Surface type classification Data masking
Sea Ice Type (SIType)	C3S SIType CDR v3.0	daily	Northern hemisphere	Sea ice density parametrization Snow climatology modification

II.5 Grid



The grid used for the data product uses the Equal-Area Scalable Earth Grid in version 2 (EASE2), which is based on a polar aspect spherical Lambert azimuthal equal-area projection and the WGS84 ellipsoid. The grid dimension is 5400 km x 5400 km with a spatial resolution 25 km, resulting in a 432 x 432 grid. The grid is centered on the geographic Pole, meaning that the Pole is located at the intersection of center cells.

II.6 Vertical Levels

Surface only.

II.7 Processing Information

II.7.1 Update Time

The data product is based on a climate data record that is not regularly updated.

II.7.2 Temporal extent of analysis and forecast stored on delivery mechanism

The product is stored since 1995-10.

II.7.3 Time averaging

Each sea ice thickness product aggregates monthly input data using averaging per grid cell.

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article:
" How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1 Other information: mean centre of Products, land mask value, missing value

In the CMEMS online system data from the latest 2 years (running window) are available via Marine Data Store

IV.2 File size

File size per product is approximately 2.5 MB.

IV.3 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

V ANNEX

The model output files contain some additional fields in addition to the variables mentioned earlier. An example output NetCDF file header is inserted below, with the additional fields in italic fonts.

netcdf file: ice_thickness_nh_ease2-250_cdr-v3p0_202004.nc

```
{
 dimensions:
   time = UNLIMITED;   // (1 currently)
   nv = 2;
   xc = 432;
   yc = 432;

variables:

   double lat(yc=432, xc=432);
     :coverage_content_type = "coordinate";
     :long_name = "latitude coordinate";
     :standard_name = "latitude";
     :units = "degrees_north";

   double lon(yc=432, xc=432);
     :long_name = "longitude coordinate";
     :standard_name = "longitude";
     :units = "degrees_east";
     :coverage_content_type = "coordinate";

   byte quality_flag(time=1, yc=432, xc=432);
     :comment = "The expert assessment on retrieval quality is only provided for grid cess with
       valid thickness retrieval";
     :coordinates = "time lat lon";
     :coverage_content_type = "qualityInformation";
     :flag_meanings = "nominal_quality intermediate_quality low_quality no_data";
     :flag_values = 0B, 1B, 2B, 3B; // byte
     :grid_mapping = "Lambert_Azimuthal_Grid";
     :long_name = "Sea Ice Thickness Quality Flag";
     :standard_name = "quality_flag";
     :units = "1";
     :valid_max = "3";
```

```
  :valid_min = "0";

float sea_ice_thickness(time=1, yc=432, xc=432);
  :ancillary_variables = "uncertainty status_flag quality_flag";
  :comment = "this field is the primary sea ice thickness estimate for this climate data
    record";
  :coordinates = "time lat lon";
  :coverage_content_type = "physicalMeasurement";
  :grid_mapping = "Lambert_Azimuthal_Grid";
  :long_name = "Sea Ice Thickness";
  :standard_name = "sea_ice_thickness";
  :units = "m";

byte status_flag(time=1, yc=432, xc=432);
  :coordinates = "time lat lon";
  :coverage_content_type = "qualityInformation";
  :flag_meanings = "nominal_retrieval no_data open_ocean satellite_pole_hole
    land_lake_landice retrieval_failed";
  :flag_values = 0B, 1B, 2B, 3B, 4B, 5B; // byte
  :grid_mapping = "Lambert_Azimuthal_Grid";
  :long_name = "Sea Ice Thickness Status Flag";
  :standard_name = "status_flag";
  :units = "1";
  :valid_max = "5";
  :valid_min = "0";

float uncertainty(time=1, yc=432, xc=432);
  :coordinates = "time lat lon";
  :coverage_content_type = "auxiliaryInformation";
  :grid_mapping = "Lambert_Azimuthal_Grid";
  :long_name = "Sea Ice Thickness Uncertainty";
  :standard_name = "sea_ice_thickness standard_error";
  :units = "m";

 double time(time=1);
   :standard_name = "time";
   :units = "seconds since 1970-01-01";
   :long_name = "Time";
   :axis = "T";
   :calendar = "standard";
   :bounds = "time_bnds":;
   :coverage_content_type = "coordinate";
```

```
double time_bnds(time=1, nv=2);
 :long_name = "Time Bounds";
  :units = "seconds since 1970-01-01";
  :coverage_content_type = "coordinate";


double xc(xc=432);
  :standard_name = "projection_x_coordinate";
  :units = "km";
  :long_name = "x coordinate of projection (eastings)";
  :coverage_content_type = "coordinate";

double yc(yc=432);
  :standard_name = "projection_y_coordinate";
  :units = "km";
  :long_name = "y coordinate of projection (northing)";
  :coverage_content_type = "coordinate";

byte Lambert_Azimuthal_Grid;
  :proj4_string = "+proj=laea +lon_0=0 +datum=WGS84 +ellps=WGS84 +lat_0=90.0";
  :semi_major_axis = 6378137.0; // double
  :longitude_of_projection_origin = 0.0; // double
  :latitude_of_projection_origin = 90.0; // double
  :false_easting = 0.0; // double
  :false_northing = 0.0; // double
  :grid_mapping_name = "lambert_azimuthal_equal_area";
  :inverse_flattening = 298.257223563; // double
```

```
// global attributes:
  :title = "Sea Ice Thickness";
  :institution = "Alfred-Wegener-Institut Helmholtz Zentrum fuer Polar und Meeresforschung";
  :source = "Altimetry: cryosat2, Snow depth: Monthly climatology based on regional merging
      of Warren Climatology with 50% reduction for first-year sea ice in the central Arctic and
      AMSR-2 based snow depth from IUP Bremen in remaining ice-covered regions, Mean Sea
      Surface: DTU15 global mean sea surface, Sea ice Concentration: C3S Sea Ice
      Concentration CDR v3p0, Sea ice type: C3S Sea Ice Type CDR v3p0";
  :platform = "CryoSat-2";
  :sensor = "SIRAL";
  :history = "20230125T013424Z (created)";
  :references = "Product User Guide and Specification (PUGS): Sea Ice Thickness Version 3,
      Algorithm Theoretical Baseline Document (ATBD): Sea Ice Thickness Version 3";
  :tracking_id = "a7e0536c-5f08-4efe-9d6a-1f60a20722a1";
  :Conventions = "CF-1.7 ACDD-1.3";
  :product_version = "3p0";
  :processing_level = "Level-3 Collated (l3c)";
  :summary = "Monthly gridded Northern Hemisphere Sea Ice Thickness Climate Data Record
      (CDR) from Envisat and CryoSat-2 satellite radar altimetry for the period October 2002 -
      April 2020";
  :topiccategory = "Oceans Climatology Meteorology Atmosphere";
  :keywords = "GCMDSK:Earth Science > Cryosphere > Sea Ice > Ice Depth/Thickness,
      GCMDSK:Earth Science > Oceans > Sea Ice > Ice Depth/Thickness, GCMDSK:Earth Science
      > Climate Indicators > Cryospheric Indicators > Ice Depth/Thickness,
      GCMDLOC:Geographic Region > Northern Hemisphere, GCMDLOC:Vertical Location > Sea
      Surface, GCMDPROV:Consortia/Institutions > AWI > Alfred Wegener Institute for Polar
      and Marine Research";
  :keywords_vocabulary = "GCMDSK:GCMD Science
      Keywords:https://gcmd.earthdata.nasa.gov/kms/concepts/concept_scheme/sciencekeyw
      ords, GCMDLOC:GCMD
      Locations:https://gcmd.earthdata.nasa.gov/kms/concepts/concept_scheme/locations,
      GCMDPROV:GCMD
      Providers:https://gcmd.earthdata.nasa.gov/kms/concepts/concept_scheme/providers";
  :id = "c3s-sit-cdr-v3p0-l3c-cryosat2-nh_25km_ease2-20200401";
  :naming_authority = "de.awi";
  :doi = "None";
  :cdm_data_type = "grid";
```

:comment = "Northern hemisphere sea ice thickness coverage is limited to the winter month between October and April due to negative effect of surface melt on the retrieval of freeboard. Please consult the Product User Guide (PUG) for more information.";

:date_created = "20230125T013424Z";

:creator_name = "Alfred-Wegener-Institut Helmholtz Zentrum fuer Polar und Meeresforschung";

:creator_type = "institution";

:creator_email = "stefan.hendricks@awi.de";

:creator_url = "http://www.awi.de/en";

:contributor_name = "Stefan Hendricks, Robert Ricker, Stephan Paul, Eero Rinne, Heidi Sallila";

:contributor_role = "PrincipalInvestigator, Author, Author, Author, Author";

:project = "European Union Copernicus Climate Change Services (C3S)";

:publisher_name = "Copernicus Climate Data Store";

:publisher_url = "https://climate.copernicus.eu/climate-data-store";

:publisher_email = "copernicus-support@ecmwf.int";

:geospatial_lat_min = "16.6239";

:geospatial_lat_max = "90.0";

:geospatial_lon_min = "-180.0";

:geospatial_lon_max = "180.0";

:geospatial_vertical_min = "0.0";

:geospatial_vertical_max = "0.0";

:spatial_resolution = "25.0 km grid spacing";

:geospatial_bounds_crs = "EPSG:6931";

:time_coverage_start = "2020-04-01T00:00:00";

:time_coverage_end = "2020-04-30T23:59:59.999999";

:time_coverage_duration = "P1M";

:time_coverage_resolution = "P1M";

:standard_name_vocabulary = "CF Standard Name Table (v36, 21 September 2016)";

:license = "C3S Copernicus License (see https://cds-dev.copernicus-climate.eu/api/v2/terms/static/20180314_Copernicus_License_V1.1.pdf)";

}